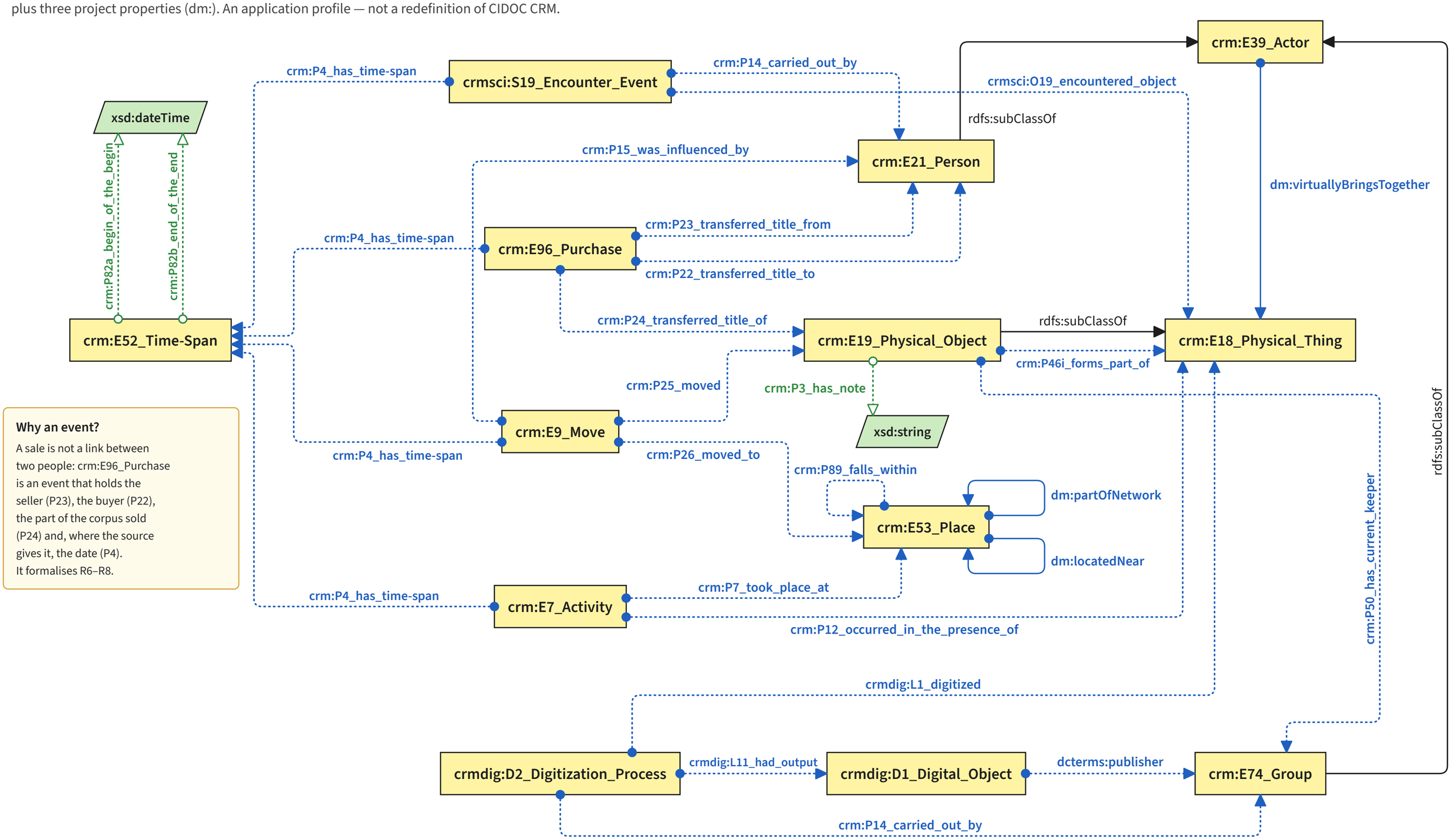
Conceptual model — core (Graffoo)

How existing ontologies are used in this project: CIDOC CRM 7.1.3, CRMsci 3.2 and CRMdig 5.0 classes and properties as used by the data, plus three project properties (dm:). An application profile — not a redefinition of CIDOC CRM.



Why an event?

A sale is not a link between two people: crm:E96_Purchase is an event that holds the seller (P23), the buyer (P22), the part of the corpus sold (P24) and, where the source gives it, the date (P4). It formalises R6–R8.

Reconnecting Dunhuang project ontology 2.1
`https://haopuur-max.github.io/reconnecting-dunhuang/ontology/`

Additional axioms (dm-ontology.ttl)

```
ObjectProperty: dm:partOfNetwork
Domain: crm:E53_Place Range: crm:E53_Place
ObjectProperty: dm:locatedNear
Characteristics: Symmetric
Domain: crm:E53_Place Range: crm:E53_Place
ObjectProperty: dm:virtuallyBringsTogether
Domain: crm:E39_Actor Range: crm:E18_Physical_Thing
```

Prefixes

crm:	http://www.cidoc-crm.org/cidoc-crm/ (7.1.3)
crmsci:	http://www.cidoc-crm.org/extensions/crmsci/ (3.2)
crmdig:	http://www.cidoc-crm.org/extensions/crmdig/ (5.0)
dcterms:	http://purl.org/dc/terms/
dm:	https://haopuur-max.github.io/reconnecting-dunhuang/ontology/
xsd:	http://www.w3.org/2001/XMLSchema#

Graffoo notation

- Class — class
- Datatype — datatype
- object property declared here
- object property used here (facility)
- data property used here (facility)
- rdfs:subClassOf

● domain end ▶ range end

Dotted arcs assert no domain or range; they show where an existing property is used. Its declared signature is on the right.

Why each reused property fits the classes it links

Declared domain → range in CIDOC CRM 7.1.3, CRMsci 3.2, CRMdig 5.0, DCMI Terms 2020. ⊑ : subclass of

Property	Declared	Used here	Valid because
crm:P4_has_time-span	E2 → E52	E96, E9, E7, S19	E96 ⊑ E8 ⊑ E7; E9 ⊑ E7; E7 ⊑ E5 ⊑ E4 ⊑ E2
			S19 ⊑ S27 ⊑ I1 ⊑ E7
crm:P7_took_place_at	E4 → E53	E7 → E53	E7 ⊑ E5 ⊑ E4
crm:P12_occurred_in_the_presence_of	E5 → E77	E7 → E18	E7 ⊑ E5; E18 ⊑ E72 ⊑ E70 ⊑ E77
crm:P14_carried_out_by	E7 → E39	S19 → E21, D2 → E74	S19 ⊑ S27 ⊑ I1 ⊑ E7; D2 ⊑ D11 ⊑ D7 ⊑ E11 ⊑ E7
			E21 ⊑ E39; E74 ⊑ E39
crm:P15_was_influenced_by	E7 → E1	E9 → E21	E9 ⊑ E7; E21 ⊑ … ⊑ E1
crm:P22_transferred_title_to	E8 → E39	E96 → E21	E96 ⊑ E8; E21 ⊑ E39
crm:P23_transferred_title_from	E8 → E39	E96 → E21	E96 ⊑ E8; E21 ⊑ E39
crm:P24_transferred_title_of	E8 → E18	E96 → E19	E96 ⊑ E8; E19 ⊑ E18
crm:P25_moved	E9 → E19	E9 → E19	direct
crm:P26_moved_to	E9 → E53	E9 → E53	direct
crm:P46i_forms_part_of	E18 → E18	E19 → E18	E19 ⊑ E18
crm:P50_has_current_keeper	E18 → E39	E19 → E74	E19 ⊑ E18; E74 ⊑ E39
crm:P89_falls_within	E53 → E53	E53 → E53	direct
crm:P3_has_note	E1 → E62	E19 → xsd:string	E19 ⊑ … ⊑ E1; E62 String as literal
crm:P82a / P82b	E52 → E61	E52 → xsd:dateTime	direct; E61 Time Primitive as literal
crmsci:O19_encountered_object	S19 → E18	S19 → E18	direct
crmdig:L1_digitized	D2 → E18	D2 → E18	direct
crmdig:L11_had_output	D7 → D1	D2 → D1	D2 ⊑ D11 ⊑ D7
dcterms:publisher	none	D1 → E74	no domain or range (range includes Agent)

Generated from reconnecting_dunhuang.ttl (683 triples) and dm-ontology.ttl. Labels, identifiers, subjects and provenance: second diagram.